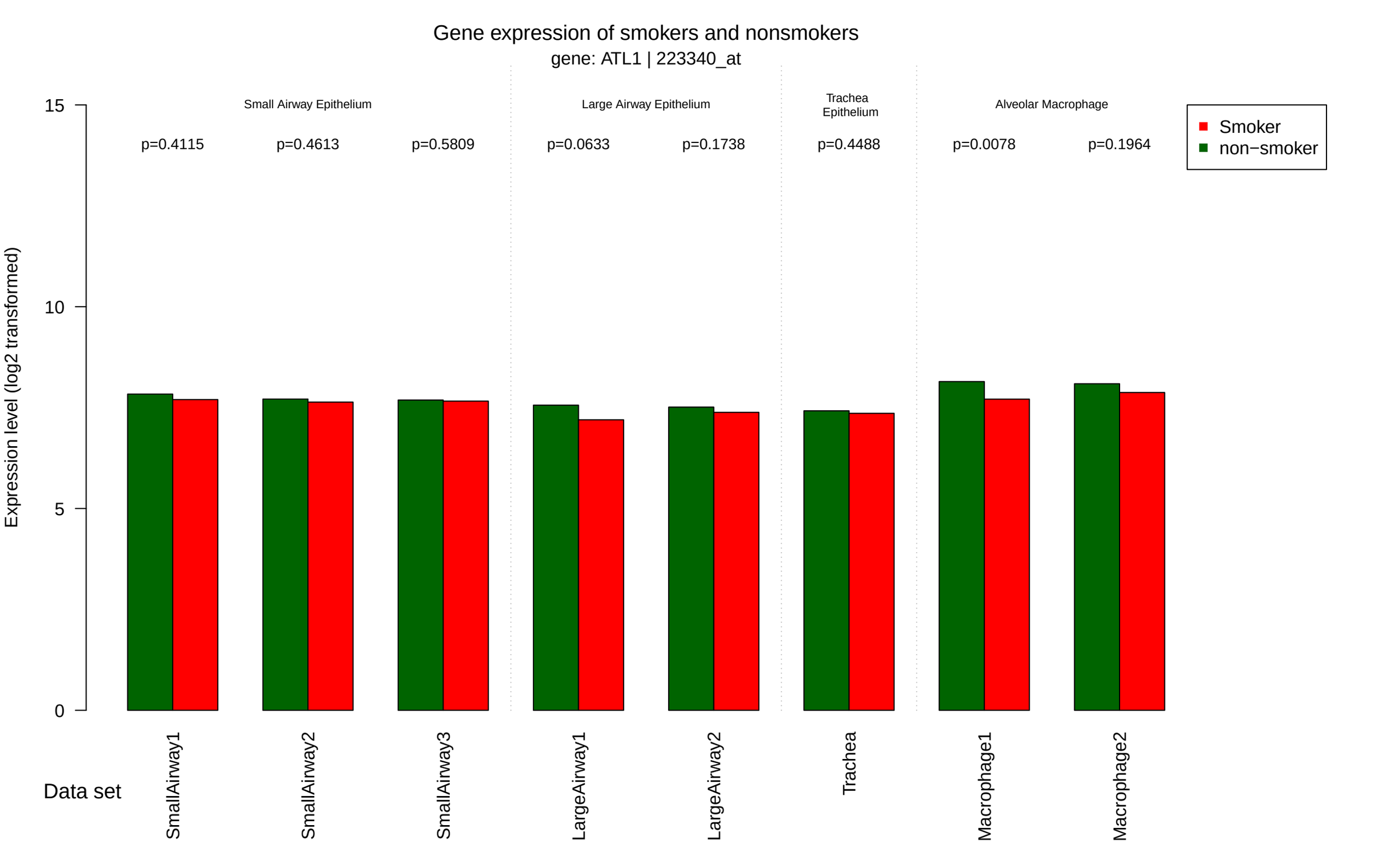
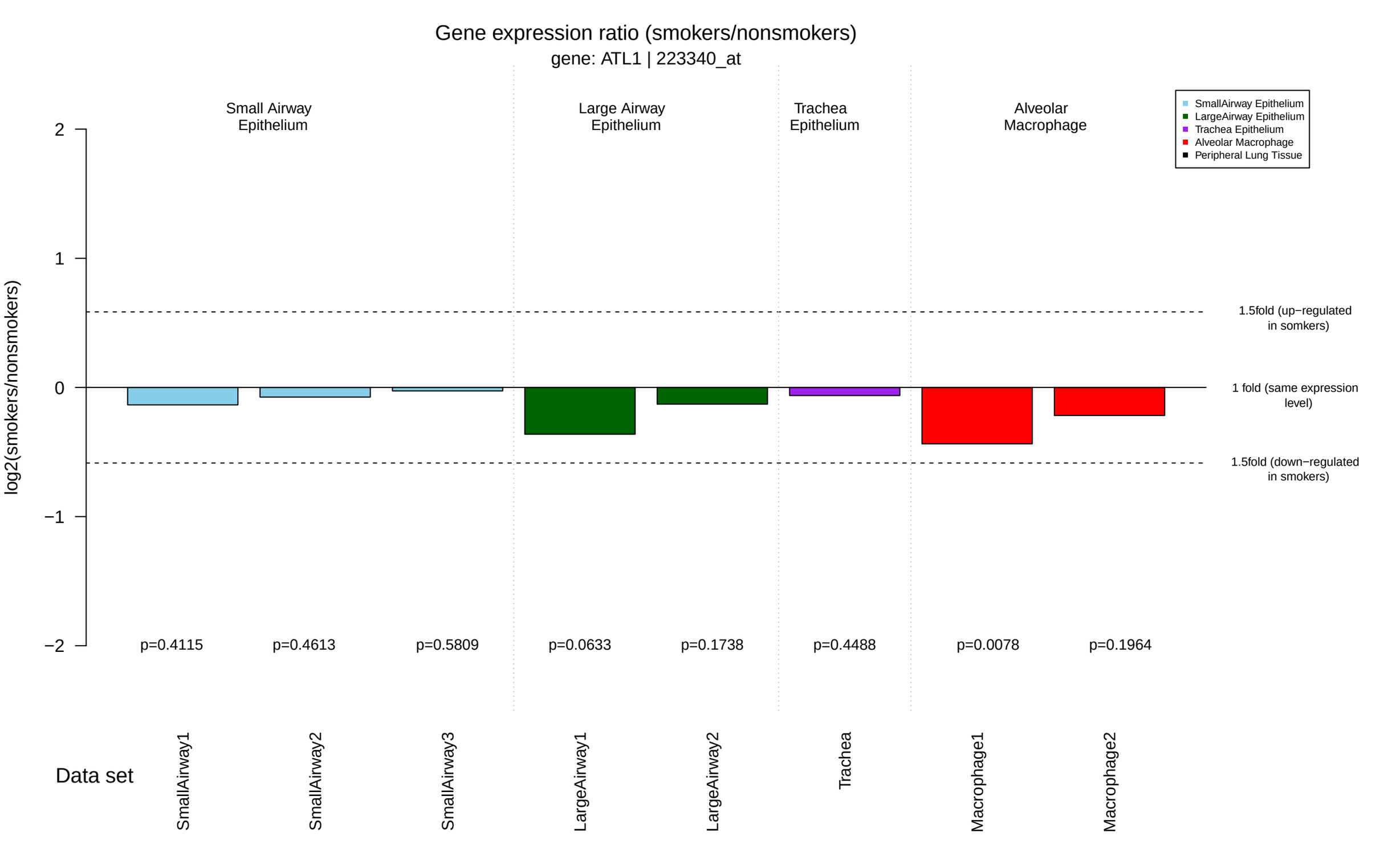
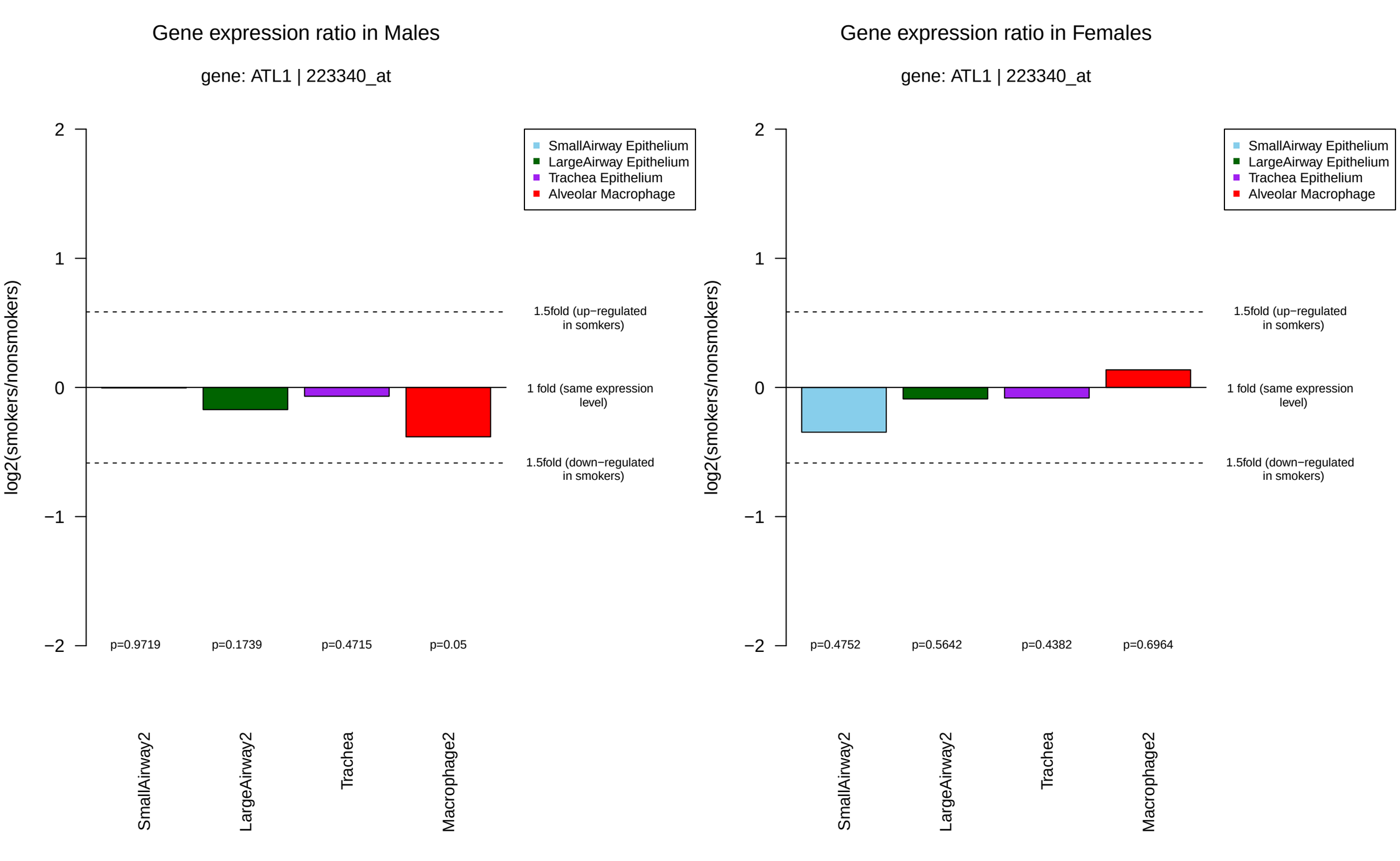
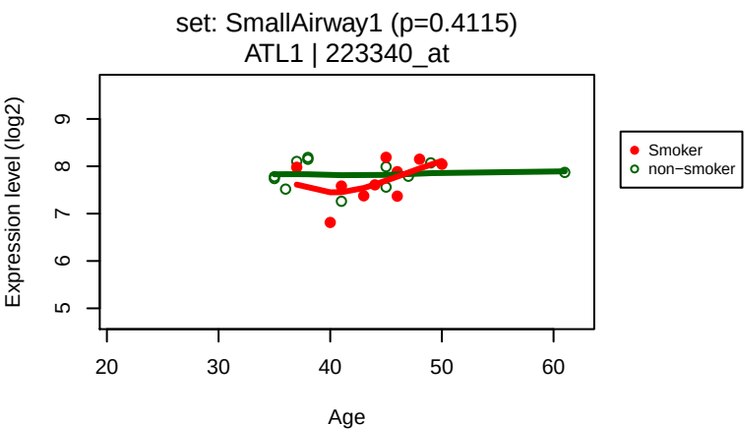
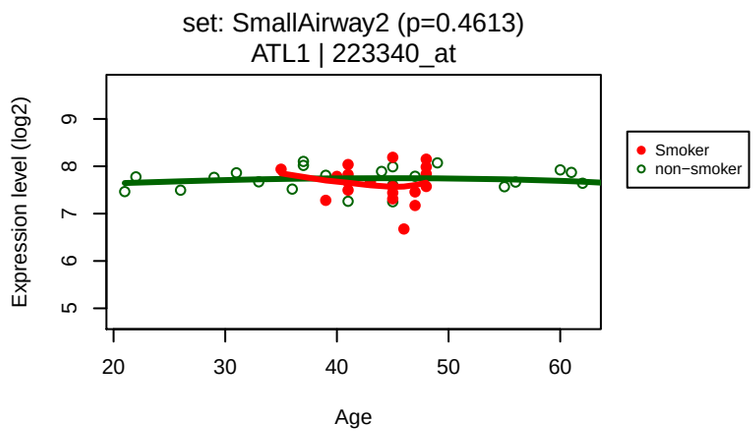
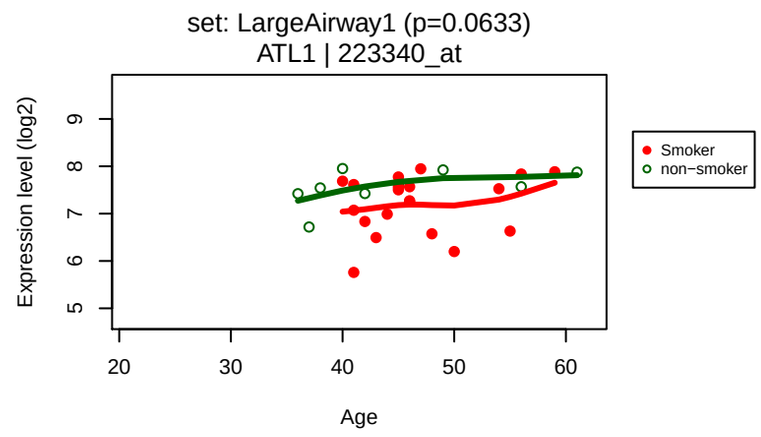

Gene expression of smokers and nonsmokers
gene: ATL1 | 223340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4115
p=0.4613
p=0.5809
p=0.0633
p=0.1738
p=0.4488
p=0.0078
p=0.1964
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

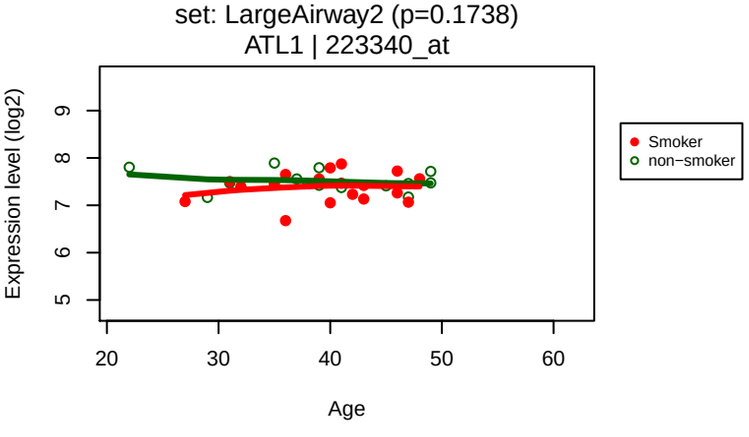

Gene expression ratio (smokers/nonsmokers)
gene: ATL1 | 223340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4115
p=0.4613
p=0.5809
p=0.0633
p=0.1738
p=0.4488
p=0.0078
p=0.1964
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

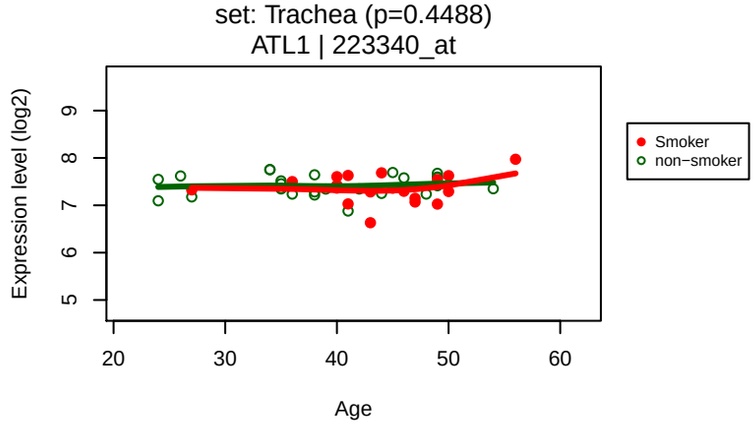

Gene expression ratio in Males
gene: ATL1 | 223340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9719
p=0.1739
p=0.4715
p=0.05
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATL1 | 223340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4752
p=0.5642
p=0.4382
p=0.6964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

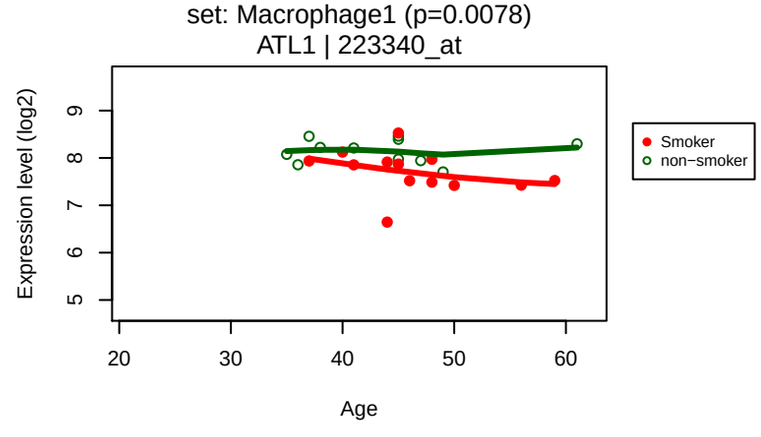

set: SmallAirway1 (p=0.4115)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker

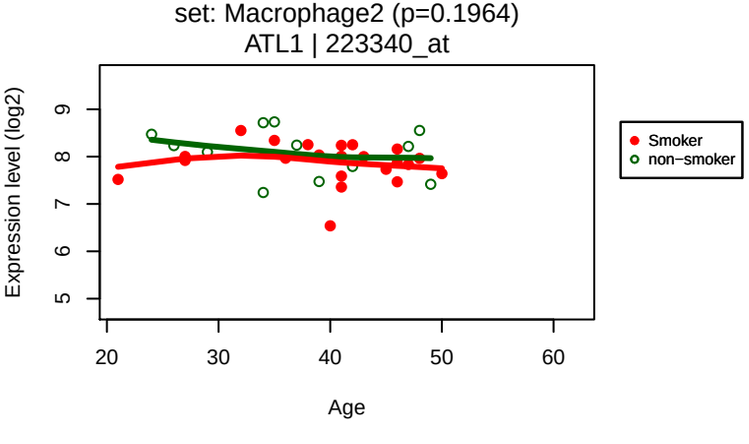

set: SmallAirway2 (p=0.4613)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker

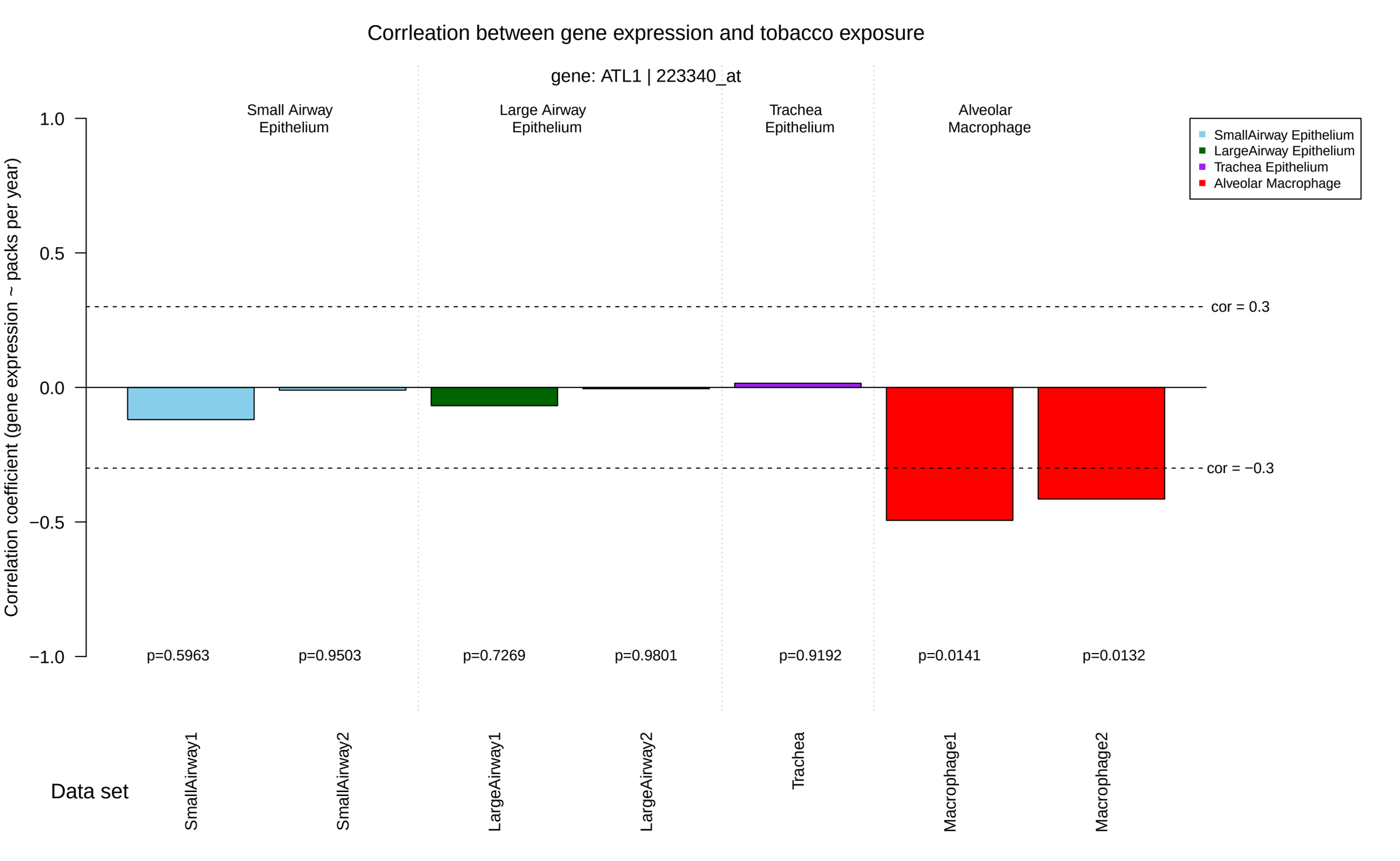

set: LargeAirway1 (p=0.0633)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1738)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4488)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0078)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1964)
ATL1 | 223340_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATL1 | 223340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5963
p=0.9503
p=0.7269
p=0.9801
p=0.9192
p=0.0141
p=0.0132
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2